

Convex-set extension via ambient interiors

`MathlibAnnex.IsometryEquiv.existsUnique_affineExtension_of_convex`

Transfers the open-connected extension theorem to convex sets whose ambient interiors are matched exactly by the isometry.

Statement

Let f be a surjective isometry from a convex subset s of a real normed space onto a subset t . Assume the ambient interior of s is nonempty and, for every x in s , x lies in $\text{interior}(s)$ if and only if $f(x)$ lies in $\text{interior}(t)$. Then f extends uniquely to an ambient real affine isometry equivalence. No separate convexity assumption on t is required by this statement.

Assumptions

Real normed vector spaces; the source s is convex with nonempty interior; f is an isometry equivalence s to t ; x is interior to s exactly when $f(x)$ is interior to t .

Conclusion

f extends uniquely to an ambient real affine isometry equivalence. No additional convexity assumption on t is stated.

Proof idea

Restrict f to an isometry between the two ambient interiors. Convexity and nonempty interior make the source interior connected, so the open-connected theorem yields an affine extension on the interiors. The interior of a convex set with nonempty interior is dense in the set; continuity therefore extends agreement to all of s . Uniqueness is inherited by restriction to the interior.

Proof steps

1. Construct an isometry equivalence between $\text{interior}(s)$ and $\text{interior}(t)$ from the bidirectional interior-membership hypothesis.
2. Apply the open-connected extension theorem to the two open interiors, using connectedness of the source interior obtained from convexity of s .
3. Identify the subtype of points of s lying in $\text{interior}(s)$ and prove it is dense in s via the convex closure-of-interior theorem.
4. Use continuity of the affine extension and of f to extend their equality from the dense interior subtype to every point of s .
5. For uniqueness, restrict any competing ambient affine isometry to the source interior and invoke the uniqueness clause from the open-connected theorem.

Main citations

`MathlibAnnex.IsometryEquiv.existsUnique_affineExtension`

The proof or construction of Convex-set extension via ambient interiors uses the project declaration “Mankiewicz extension on open connected domains” at the indicated step.

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The proof or construction of Convex-set extension via ambient interiors uses the project declaration “Restriction of an isometry to ambient interiors” at the indicated step.

Lean source signature (exact)

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theorem existsUnique_affineExtension_of_convex
  {s : Set E} {t : Set F} (f : s  $\approx_i$  t)
  (hs : Convex  $\mathbb{R}$  s) (hsint : (interior s).Nonempty)
  (hfi :  $\forall x : s,$ 
     $(x : E) \in \text{interior } s \Leftrightarrow ((f \ x : t) : F) \in \text{interior } t$ ) :
   $\exists! A : E \approx^{a_i}[\mathbb{R}] F,$ 
   $\forall x : s, A \ (x : E) = ((f \ x : t) : F)$ 
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[Read exact MathlibAnnex v0.4.0 source](#)

Exact Card identity

Stable Card ID: 45c4db7ff375613be920f48ef4a7427063daaa3084fa98c5dd90c595522e7428

Card SHA-256: e9a5e15ee95ad76de6435c37a4fee43c4bdda17836534bf186964da0c62a9c4f

Approved exposition SHA-256: e3280713e4d5094b18795c83a17dda49595e5719bd61fc723d19165618fcbd63

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